Student 19 **Mrudhula K** 192524335RC 20 / 20 submitted

Q1. Union of Sets

Score: 0.00 TC: 0/1 passed

CODE

```
A={1,2,3}
B={3,4,5}
print(A|B)
```

OUTPUT

```
{1, 2, 3, 4, 5}
```

Q2. Intersection

Score: 0.00 TC: 0/1 passed

CODE

```
A={1,2,3}
B={2,3,4}
c=A&B
print(c)
```

OUTPUT

```
{2, 3}
```

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A={1,2,3}
B={2,3,4}
print(A.difference(B))
```

OUTPUT

```
{1}
```

Q4. Symmetric Difference

Score: 0.00 TC: 0/1 passed

CODE

```
a={1,2,3}
b={2,3,4}
print(a^b)
```

OUTPUT

```
{1, 4}
```

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A={2,3}
B={1,2,3,4}
if A.issubset(B):
    print("Subset")
else:
    print("Not a Subset")
```

OUTPUT

```
Subset
```

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A={1,2,3,4}
B={2,3}
if A.issuperset(B):
    print("Superset")
else:
    print("Not a Superset")
```

OUTPUT

```
Superset
```

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A={1,2}
B={3,4}
if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not a Disjoint")
```

OUTPUT

```
Disjoint
```

Q8. Remove Duplicates

Score: 0.00 TC: 0/1 passed

CODE

```
arr=[1,2,2,3,3,4]
print(set(arr))
```

OUTPUT

```
{1, 2, 3, 4}
```

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A={1,2,3}
B={2,3,4}
C={2,5}
print(A&B&C)
```

OUTPUT

```
{2}
```

Q10. Missing Numbers

Score: 0.00 TC: 0/1 passed

CODE

```
A=set(map(int,input().split()))
B=set(range(1,11))
print(B-A)
```

INPUT

```
1 2 4 5 7 10
```

OUTPUT

```
{8, 9, 3, 6}
```

Q11. First Repeated

Score: 5.00 **TC: 1/1 passed**

CODE

```
arr=list(map(int,input().split()))
seen=set()
for i in arr:
    if i in seen:
        print(i)
        break
    seen.add(i)
```

INPUT

```
10 20 30 20 40
```

OUTPUT

```
20
```

Q12. Duplicate Elements

Score: 0.00 **TC: 0/1 passed**

CODE

```
arr=[10,20,30,20,10]
duplicates=set()
seen=set()
for i in arr:
    if i in seen:
        duplicates.add(i)
    else:
        seen.add(i)
print(duplicates)
```

INPUT

```
10 20 30 20 10
```

OUTPUT

```
{10, 20}
```

Q13. Compare Lists

Score: 0.00 **TC: 0/1 passed**

CODE

```
list1=list(map(int,input().split()))
list2=list(map(int,input().split()))
print(set(list1)==set(list2))
```

INPUT

```
1 2 2 3
2 3 2 1
```

OUTPUT

```
True
```

Q14. Duplicate Tuples

Score: 0.00 **TC: 0/1 passed**

CODE

```
arr=[(1,2),(1,2),(3,4)]
result=list(set(arr))
print(result)
```

OUTPUT

```
[(1, 2), (3, 4)]
```

Q15. Pair Sum

Score: 0.00 **TC: 0/1 passed**

CODE

```
arr=list(map(int,input().split()))
target=int(input())
pairs=[]
for i in range(len(arr)):
    for j in range(i+1, len(arr)):
        if arr[i]+arr[j]==target:
            pairs.append((arr[i],arr[j]))
print(*pairs,sep=",")
```

INPUT

```
2 4 6 8 10
12
```

OUTPUT

```
(2, 10),(4, 8)
```

Q16. Unique Characters

Score: 5.00 **TC: 1/1 passed**

CODE

```
s=input()
unique=set(s)
print(len(unique))
```

INPUT

```
Programming
```

OUTPUT

```
8
```

Q17. Unique Words

Score: 0.00 **TC: 0/1 passed**

CODE

```
sentence=input()
words=sentence.split()
print(set(words))
```

INPUT

```
python is easy python
```

OUTPUT

```
{'easy', 'python', 'is'}
```

Q18. Exactly One Set

Score: 0.00 **TC: 0/1 passed**

CODE

```
A={1,2,3}
B={3,4,5}
C={5,6,7}
result=(A-B-C)|(B-A-C)|(C-A-B)
print(result)
```

OUTPUT

```
{1, 2, 4, 6, 7}
```

Q19. Menu Driven

CODE

```
A=set(map(int,input().split()))
B=set(map(int,input().split()))
choice=int(input())
if choice==1:
    print("union:", A|B)
elif choice==2:
    print("intersection:", A&B)
elif choice==3:
    print("difference:", A-B)
elif choice==4:
    print("symmetric:", A^B)
elif choice==5:
    print("exit")
else:
    print("invalid choice")
```

INPUT

```
1 2 3 4
3 4 5 6
1
```

OUTPUT

```
union: {1, 2, 3, 4, 5, 6}
```

Q20. Student Database

CODE

```
students=set()
name=input()
students.add(name)
print("student added successfully\n")
name=input()
students.add(name)
print("student added successfully\n")
print("students:\n", students)
name=input()
students.remove(name)
print("student removed successfully")
name=input()
if name in students:
        print("found")
print("no of students:", len(students))
```

INPUT

```
ravi
priya
ravi
ravi
```

OUTPUT

```
Security Error: Disallowed code pattern detected.
```